The amplitude to detect the particle at $|x_2\rangle$ is just the inner product of $|\Psi(t_2)\rangle$ with $|x_2\rangle$. Its value is

$$C_{1,2} = \langle x_2 | e^{-iH(t_2 - t_1)} | x_1 \rangle. \qquad (9.21)$$

In other words, the amplitude to go from $x_1$ to $x_2$ over the time interval $t_2 - t_1$ is constructed by sandwiching $e^{-iH(t2-t1)}$ between the initial and final positions. To simplify the formula, let's define $t_2 - t_1$ to be $t$. Then the amplitude is

$$C_{1,2} = \langle x_2 | e^{-iHt} | x_1 \rangle. \qquad (9.22)$$

Now, let's break the time interval $t$ into two smaller intervals of size $t/2$ (see Fig. 9.6). The operator $e^{-iHt}$ can be written as the product of two operators:

$$e^{-iHt} = e^{-iHt/2} e^{-iHt/2}. \qquad (9.23)$$

By inserting the identity operator in the form

$$I = \int dx |x\rangle\langle x|, \qquad (9.24)$$

we can rewrite the amplitude as

$$C_{1,2} = \int dx \langle x_2 | e^{-iHt/2} | x \rangle \langle x | e^{-iHt/2} | x_1 \rangle. \qquad (9.25)$$

This form of the equation looks more complicated, but has a very interesting interpretation. Let me put it in words. The amplitude to get from $x_1$ to $x_2$ over time interval $t$ is an integral over an intermediate position $x$. The integrand is the amplitude to go from $x_1$ to $x$ over the time interval $t/2$ *multiplied by* the amplitude to go from $x$ to $x_2$ over another time interval $t/2$.